Quantum Readout from Observer-Relative Lossy Descent

Tau Core Foundation Paper VI: A finite conditional quantum terminal

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Abstract

We ask which additional structures are required for an observer-relative, information-losing descent from an atemporal morphological body to support a quantum terminal rather than ordinary classical coarse graining. First, a finite countermodel proves that resolution loss, non-injectivity and stable record classes alone imply neither complex amplitudes nor interference, Born probabilities, entanglement or no-signalling. We then state a finite conditional source packet. A positive quotient metric and a compatible nondegenerate two-form construct a complex Hilbert carrier when the associated endomorphism has one common complex scale. A commuting self-adjoint generator gives unitary transport; positive normalized effects give the trace Born rule; and an explicitly selected tensor product gives local-CPTP no-signalling.

Inside an adopted enriched Tau completion, these structures are represented on one normalized complex matrix block with a common action unit. We prove finite record-separation and tensor-factorization criteria, distinguish exact logical recovery from dephasing, and give an approximate-recovery bound. The standard free algebraic-QFT and perturbative causal-QFT constructions are used only as conditional fixed-background limits. None of these standard quantum results is presented as a Tau-specific deviation.

The proposed discriminator instead compares two independently reconstructed quantities on the same occupied record pair: a normalized body-action separation and its Uhlmann chord. The completion requires them to agree, with a certified recovery-error envelope. A public-data audit finds no carrier on which the action-side quantity is independently observable, so the discriminator is not yet executable. We also prove that the present narrow base–seed reduct does not select the full joint source packet. Consequently the paper establishes a mathematically explicit conditional quantum terminal, not a quantum theory derived from the unrestricted physical parent law; it does not derive the SI value of $\hbar$, physical outcome selection, nonperturbative QFT, quantized geometry or the Standard Model.

Dependency and ownership. Inputs are: the atemporal body and observer-access architecture (Paper I), typed finite source support (Paper II), recovery-correctable record transport (Paper III), the common-source occupation boundary (Paper IV), and stable observer quantization plus a calibrated terminal clock (Paper V). This paper owns the loss-only no-go, the finite metric–symplectic quantum completion, Born/interference and no-signalling terminals, and the quantum/classical status ledger. It also owns the recovery-based factor selector, its approximate error bound and the central-chirality terminal-blindness theorem. It neither re-proves nor silently strengthens its inputs. The downstream scalar-15/type-II mass and RG specialization is owned by Paper VII-C; it does not replace the observer quantizer or the quantum source packet proved here.

Atemporal parent-realization terminology. Because the Tau parent is atemporal, phrases such as "Nature selects" do not denote an external agent or a later choice. Their precise meaning is the internal physical implication

$$(B_\tau^{\mathrm{phys}}, s_U, \mathcal{L}_{\mathrm{parent}}) \Longrightarrow \mathfrak{P},$$

together with occupation of the support of the resulting packet $\mathfrak{P}$. A theorem inside a declared completion proves only class-relative realization. It does not prove this unrestricted base–seed arrow, and agreement of finitely many terminal readouts cannot by itself establish ambient-source exhaustivity. Current-facing statements therefore use *parent-law realization* and *physical occupation*; any retained legacy selection language is shorthand for this same distinction.

1 Inherited proof hierarchy

Paper VI is a consolidation and terminal-recovery paper, not a restart of the quantum branch. A corpus-wide audit of the theory hub and Papers I–V gives the following inherited ladder.

Level	Inherited result	Status used here
Algebraic	A full-role operator square $C(v)^2 = K_*(v,v)I$ polarizes to the Clifford relations	Exact implication
Representation	Seven P3 generators admit the finite packet $\mathrm{Cl}_7(\mathbb{C}) \simeq M_8(\mathbb{C}) \oplus M_8(\mathbb{C})$	Constructed conditional representation
Completion	The graph-normal P3 handoff realizes the operator square with one action unit and zero body Schur remainder	Adopted minimal post-body completion
Record/terminal	Full S4 effects and the S5 square-root terminal supply the sharp spectral instruments; the matrix block supplies metric, phase, unitary and Born structures	Conditional recovery inside the enriched MVP
Field limit	Globally hyperbolic descent, Hadamard preparation and local causal factorization supply free and perturbative interacting QFT structure	Conditional standard recovery; interaction content and nonperturbative limit open
Physical source	The current narrow parent has no typed full-role incidence into the squared-Hodge carrier	Exact current-source no-entailment; unrestricted enriched-packet occupation open

Table 1: Inherited proof hierarchy. The open top-level source arrow does not erase the lower-level theorems, while the lower-level constructions do not prove unrestricted physical occupation.

This hierarchy fixes the interpretation of the paper. The mathematical P3 and quantum-terminal machinery is already available at several levels. The remaining source question is the typed incidence

$$\iota_{P3} : (E_{P3}, K_*) \longrightarrow (R_{\mathrm{common}}, h_{\mathrm{common}}), \tag{1}$$

selected before the handoff. Existing EACER support is rank two; the rank-nine seed-normal and endpoint/context carriers lack the required typed map; and the common Clifford envelope is an operator target rather than the source incidence. Paper VI uses the adopted completion for forward terminal theorems and keeps Eq. (1) open at the unrestricted physical parent-law level.

2 Question, architecture and claim boundary

Let M_τ denote a stabilized atemporal morphological body generated by the physically enriched Tau base and a universal seed. For an observer carrier O, source region S and context c, the

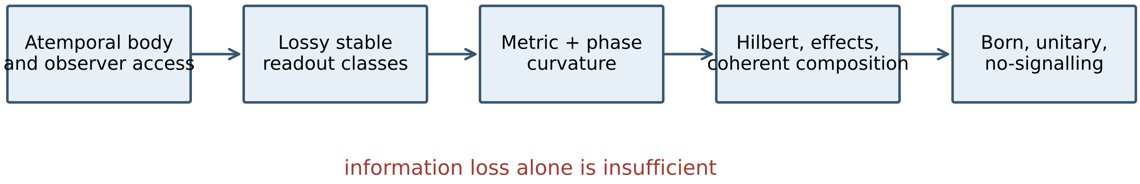

Figure 1: The Paper-VI descent. Finite resolution supplies stable classes, whereas complex phase geometry, effects and composition require additional source-owned structure. Dashed implications are conditional.

accessible terminal information is not the body itself. It is obtained through a body-conditioned relation and a stable finite-resolution map,

$$B_\tau^{\rm phys} + s_U \longrightarrow M_\tau \longrightarrow \mathcal{C}_{OS|c}(M_\tau) \xrightarrow{Q_{OS|c}} \mathcal{R}_{OS|c}. \tag{2}$$

Here $\mathcal{C}_{OS|c}$ is not a separate substance. It is the observer–source realization of relations already conditioned by the body. The stable quantizer $Q_{OS|c}$ defines observer-relative readout classes. Equality of classes, rather than a non-transitive distance threshold, defines the operational equivalence relation

$$x \sim_{OS|c} y \quad \Longleftrightarrow \quad Q_{OS|c}(x) = Q_{OS|c}(y). \tag{3}$$

The central question is finite and deliberately narrow:

> Which additional source-owned structures turn the quotient $\mathcal{C}_{OS|c}/\sim_{OS|c}$ into a complex quantum carrier, and which familiar quantum laws then follow without further fitted gains?

Figure 1 displays the complete logical direction. In particular, terminal observations constrain a frozen forward construction; they do not ontologically build M_τ backwards.

3 Relation to established structures and novelty boundary

The terminal mathematics used below is standard once its inputs are granted. Stinespring/Kraus theory supplies completely positive channels [1, 2]; Lindblad dynamics supplies a standard decoherence limit [3]; Uhlmann geometry supplies mixed-state purification distance [4]; Gleason-type results constrain noncontextual probability measures [5]; and quantum error correction supplies exact recoverability criteria [6, 7]. Quantum-geometric response and speed-limit relations are likewise established structures [8, 9]. Paper VI does not claim these results as new consequences of information loss.

The novelty claim is narrower and conditional. Tau Core asks whether one body-derived, observer-relative source packet can own the metric, phase, action unit, effects, subsystem composition and recovery domain together, rather than importing them as unrelated terminal axioms. The paper gives finite sufficient conditions, countermodels to weaker packets and one proposed cross-domain discriminator. It does not prove that the physical base–seed system occupies the sufficient packet. This distinction separates the paper's source-architecture proposal from its use of standard quantum and QFT theorems.

4 The common accessibility quotient

Definition 1 (Stable observer quantizer)**.** *A stable observer quantizer is an idempotent measurable map $Q_{OS|c} : \mathcal{C}_{OS|c} \to \mathcal{R}_{OS|c}$ whose fibres form a partition on the occupied source family and remain invariant under the declared record-tolerance class. Its finite capacity is the number, measure or effective dimension of occupied distinguishable fibres.*

This definition retains observer dependence. Different carriers, source supports or contexts may induce different partitions. Microscopic effective discreteness and macroscopic effective continuity can therefore coexist: a fine carrier resolves fewer equivalence identifications, while redundant compatible records make a dense family of transitions operationally smooth.

For an ordered source path γ, time quantization is the smallest ordered transition whose endpoints occupy distinct stable fibres. It is not a universal atom of time unless independence from O, S, c is proved. Likewise, multiple parent configurations in one ordinary class are merely unresolved; they are not yet a coherent superposition.

5 Loss-only no-go

Proposition 1 (Loss-only no-go)**.** *Finite capacity, non-injectivity and a stable ordered quotient do not imply a complex Hilbert space, phase-sensitive composition, the Born rule, entanglement or no-signalling.*

Proof. Take the reported state space to be the classical bit simplex Δ_2. One completion is Δ_2 itself with stochastic maps. A second completion is the qubit density-operator space followed by complete dephasing in a fixed basis,

$$\mathcal{D}(\rho) = \mathrm{diag}(\rho) \in \Delta_2. \tag{4}$$

Both have exactly the same resolved quotient, occupied probability states and finite outcome alphabet. Before dephasing, the quantum completion also has off-diagonal phase and interference, whereas the classical completion does not. Adding the same order to the two reported outcomes changes neither construction. Hence the common loss/quotient data cannot select any of the listed quantum properties. □

This no-go protects the program from an otherwise tempting circularity. It also explains why observer-dependent quantization is necessary but not sufficient. Figure 2 summarizes the fork.

6 Frozen quantum source packet

We now state the smallest packet used in the positive construction. Every item must arise from one frozen post-body source; the packet may not be assembled by fitting terminal residuals.

Definition 2 (Quantum source packet)**.** *On an occupied finite-dimensional horizontal tangent space V_O, let*

$$\mathfrak{P}_Q = (g, \theta, \omega, U_A, K, \mathfrak{E}, \otimes), \qquad \omega = d\theta, \tag{5}$$

where:

(Q1) g is the positive quotient of the body Hessian;

(Q2) θ is a source-owned ordered-action one-form and ω its nondegenerate curvature on the quantum leaf;

(Q3) $A = g^{-1}\omega$ obeys $A^2 = -\kappa^2 I$ for one $\kappa > 0$;

(Q4) $U_A > 0$ is the common action unit, with no block-dependent gain;

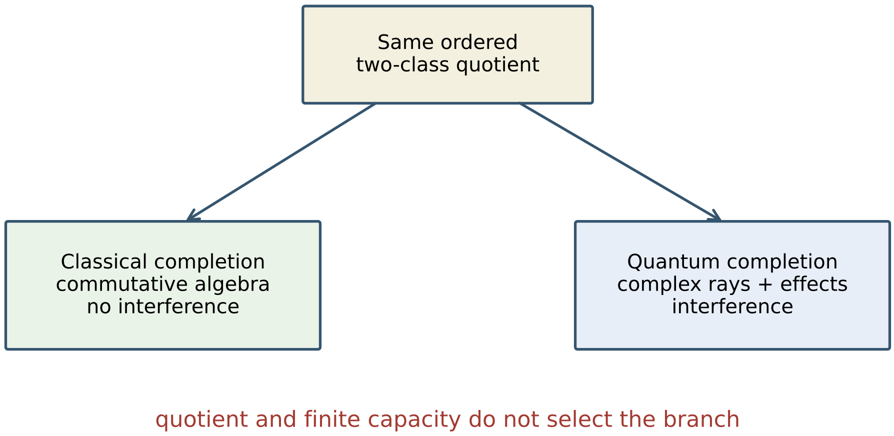

Figure 2: The same finite observer quotient admits a classical simplex and a complex quantum carrier. The distinguishing datum is phase-capable coherent composition, not information loss by itself.

(Q5) *K is g-self-adjoint and commutes with $J = A/\kappa$;*

(Q6) *$\mathfrak{E}$ is a normalized positive effect family represented by the source-owned trace state;*

(Q7) *$\otimes$ is a declared composition law on independently accessible occupied subsystems, and local deterministic operations are CPTP.*

Items (Q1)–(Q3) are the metric–symplectic completion. Items (Q4)–(Q6) select dynamics and readout statistics. Item (Q7) is logically independent: no-signalling cannot be proved before a subsystem composition is defined.

7 Existing Tau source realization

The packet can now be evaluated against earlier Tau source results rather than left as a free list. Inside the declared source-irredundant enriched MVP, the existing occupation theorem supplies an occupied coherent P3 representation and a faithful record edge. The adopted graph-normal P3 handoff has stationary operator coordinate

$$S_{P3} = c_K(e_{P3}) \tag{6}$$

in the complex Clifford carrier, with one action unit $\mathcal{A}_*$ and the normalized trace. Its finite source-minimal block ledger is

$$\mathrm{Cl}_7(\mathbb{C}) \simeq M_8(\mathbb{C}) \oplus M_8(\mathbb{C}). \tag{7}$$

Restrict to one occupied simple block $\mathcal{A}_Q = M_n(\mathbb{C})$ and let $\tau_n = \mathrm{Tr}\,/n$. On $\mathcal{A}_Q$ define

$$h(X,Y) = \mathcal{A}_*\tau_n(X^\dagger Y), \quad g = \mathrm{Re}\, h, \quad JX = iX, \quad \omega(X,Y) = g(JX,Y). \tag{8}$$

Because the matrix block is an affine vector space, the canonical linear potential

$$\theta_X(Y) = \frac{1}{2}\omega(X,Y) \tag{9}$$

satisfies $d\theta = \omega$. Thus the construction supplies the one-form in (Q2), not only its curvature.

Theorem 1 (Existing-source single-system packet)**.** *Within the adopted enriched MVP/P3 handoff, Eqs.* (8)*–*(9) *construct (Q1)–(Q4) on each occupied simple block in one source action unit. Every handoff-selected Hermitian generator and positive normalized effect family is then represented in the canonical forms required by (Q5)–(Q6). No independent metric, phase or Born gain is required; the physical choice of generator and occupied effects remains source data.*

Proof. Faithfulness of τ_n makes g positive. Scalar multiplication gives $J^2 = -I$ and preserves g, so $A = g^{-1}\omega = J$ and (Q3) holds with $\kappa = 1$. Antisymmetry of ω follows from Hermitian symmetry, Eq. (9) supplies $d\theta = \omega$, and nondegeneracy follows by choosing $Y = JX$: $\omega(X, JX) = g(X, X)$. For $H = H^\dagger$, left multiplication $K_H(X) = HX$ is self-adjoint for h and commutes with J, so $\exp(tJK_H/\mathcal{A}_*)$ is unitary. Positive matrices r, E with $\tau_n(r) = 1$ and $0 \le E \le I$ define the standard terminal density $\rho = r/n$, for which

$$\operatorname{Tr} \rho = 1, \qquad p(E|\rho) = \tau_n(rE) = \operatorname{Tr}(\rho E). \tag{10}$$

Positivity, normalization and coherent interference follow from this same pairing. All structures are restrictions of the trace-normalized handoff. □

This consolidates earlier results but has two sharp boundaries. First, the unchanged explicit parent does not select the required full-role squared-Hodge incidence: its retained phase carrier is only $\mathbb{CP}^0$ and its phase/body mixing block is exactly zero. The theorem therefore remains conditional on the already adopted P3 handoff. Second, $M_8(\mathbb{C})$ is abstractly isomorphic to $M_2(\mathbb{C})^{\otimes 3}$, but the simple algebra and trace do not select a unique tensor factorization. Inner automorphisms preserve the single-system packet while moving candidate local factors. Thus (Q7), entanglement locality and subsystem no-signalling still require a source-owned commuting-factor ledger.

The remaining handoff condition has a scalar record formulation. Let $C : E_{P3} \to \mathcal{A}_{\mathrm{occ}}^{\mathrm{sa}}$ be a linear typed response and let Φ_{sep} be a source-owned separating family of normalized positive functionals.

Theorem 2 (Separating-record selector)**.** *If, for every v and every $\varphi \in \Phi_{\mathrm{sep}}$,*

$$\varphi(C(v)^2) = K_*(v, v), \tag{11}$$

then $C(v)^2 = K_*(v, v)I$, polarization gives the Clifford relations, and the strictly convex complete-mediation theorem selects the graph-normal handoff without a bypass or new gain.

Proof. The self-adjoint difference $D_v = C(v)^2 - K_*(v, v)I$ is annihilated by every member of a separating family, so $D_v = 0$. Applying this equality to $u+v$ and subtracting the diagonal instances gives $C(u)C(v) + C(v)C(u) = 2K_*(u, v)I$. Strict convexity then makes $S_{P3} = C(e_{P3})$ the unique mediated zero locus. □

This reduces the unrestricted parent-handoff question to a finite scalar source law, but does not assume it has been observed. Normalized-trace agreement alone is insufficient: $D(v) = \sqrt{2}\,\operatorname{diag}(v_1, v_2)$ has the correct averaged quadratic trace, while its two pure diagonal records return different squares. Paper III supplies an operational informational-completeness criterion for Φ_{sep}; it does not prove physical occupation of Eq. (11).

7.1 Finite P3 separating-record rank

The separating requirement can be evaluated without introducing another operator family. On one irreducible P3 block let $\gamma_1, \ldots, \gamma_7$ be Hermitian Clifford generators and, for an ordered subset $A = \{a_1 < \cdots < a_k\}$, set

$$W_A = i^{k(k-1)/2}\gamma_{a_1} \cdots \gamma_{a_k}. \tag{12}$$

Each W_A is Hermitian, so $\rho \mapsto \operatorname{Tr}(\rho W_A)$ is a real record coordinate.

Theorem 3 (P3 separating-record rank)**.** *On one irreducible $M_8(\mathbb{C})$ block, the real ranks of the cumulative record families with Clifford grades at most $0, 1, 2, 3$ are*

$$\boxed{1, \quad 8, \quad 29, \quad 64}, \tag{13}$$

and their unresolved self-adjoint kernel dimensions are respectively

$$\boxed{63, \quad 56, \quad 35, \quad 0}. \tag{14}$$

Consequently, the unit and seven primitive P3 records are not separating, and even complete binary-word occupation is insufficient. Complete occupation through ternary words is separating.

Proof. The grade multiplicities are $\binom{7}{0} = 1$, $\binom{7}{1} = 7$, $\binom{7}{2} = 21$ and $\binom{7}{3} = 35$. In an irreducible odd-Clifford block the volume word is scalar and Hodge duality identifies grades $7-k$ and k. The words through grade three are orthogonal under normalized trace and total $1+7+21+35 = 64 = \dim_{\mathbb{R}} M_8(\mathbb{C})_{\mathrm{sa}}$; hence they form a real basis. Truncation gives Eq. (13), and subtraction from 64 gives Eq. (14). □

This closes the rank question but not the source question. Algebraic generation of W_A does not make it a physically occupied stable record, and Frobenius copying of an occupied record does not make the record family informationally complete. The remaining decision is finite: whether the base–seed record law occupies all $|A| \leq 3$ words with the common scalar action in Eq. (11). No grade-four-or-higher record is needed on this block.

7.2 Canonical spectral occupation and the Frobenius no-go

The rank theorem also identifies a canonical positive record realization. Every Hermitian Clifford word is an involution, so define

$$E_A^{\pm} = \frac{1}{2}(I \pm W_A). \tag{15}$$

These are orthogonal positive effects and $E_A^+ + E_A^- = I$. Different noncommuting words label different binary contexts, not one joint pointer partition.

Theorem 4 (Ternary spectral-record occupation)**.** *If the physical instrument is spectrally closed on every occupied grade-at-most-three Clifford involution, with the common trace/effect pairing and no context-dependent gain, the resulting 63 nontrivial binary instruments form a separating positive record atlas. Together with Eq. (11), this closes the record hypothesis of Theorem 2.*

Proof. For every state ρ,

$$p_A^+ - p_A^- = \operatorname{Tr} \rho(E_A^+ - E_A^-) = \operatorname{Tr}(\rho W_A). \tag{16}$$

The words through grade three form a real basis by Theorem 3. Equal binary outcome probabilities therefore imply equal expectations on a basis and hence equal states. □

This spectral closure is not a consequence of the current Frobenius record law. A countermodel retains the same matrix algebra and all Clifford words, but makes only the eight projectors of one distinguished diagonal basis into physical pointer records. The coproduct $|k\rangle \mapsto |k\rangle \otimes |k\rangle$ remains exact, special and coassociative, while the record span has rank only eight and misses every off-diagonal state direction. Thus redundant pointer copying and algebraic word generation do not imply complete spectral-record occupation. The remaining source law is finite and explicit:

$$W = W^\dagger, \quad W^2 = I, \quad |A| \leq 3 \quad \Longrightarrow \quad \{(I+W)/2, (I-W)/2\} \subset \mathcal{E}_{\mathrm{rec}}^{\mathrm{occ}}. \tag{17}$$

Its selection by the physical base–seed instrument action remains open.

The previously established effect/instrument branch closes this implication inside the adopted enriched MVP. If S4 is the full effect interval $\{E \in \mathcal{A}_{\mathrm{occ}}^{\mathrm{sa}} : 0 \leq E \leq I\}$ and S5 assigns the apparatus-free square-root instrument to every sourced effect, then Eq. (15) already belongs to S4 and

$$\mathcal{I}_A^{\pm}(\rho) = E_A^{\pm}\rho E_A^{\pm} \tag{18}$$

is its sharp S5 terminal. The two maps have a trace-preserving sum and each selected outcome is immediately repeatable. Therefore

$$\boxed{\text{P3HAND-D1} + \text{full S4 effect interval} + \text{S5 terminal} \Longrightarrow \text{ternary spectral-record occupation.}} \tag{19}$$

No further instrument gain or apparatus orientation is needed. This is an existing-source theorem inside the full enriched packet, not a derivation from the unchanged narrow parent: unrestricted realization and occupation of P3HAND-D1 and the complete S4/S5 packet remain open.

There is a strict directionality boundary. This argument assumes P3HAND-D1 to obtain the involutions W_A and therefore cannot be used backwards in Theorem 2 to select that same handoff. It closes the record atlas after adoption. A noncircular P3 selector would require the separating scalar evaluations on a pre-handoff source carrier defined without the Clifford representation.

8 Metric–symplectic quantum completion

Theorem 5 (Metric–symplectic quantum completion)**.** *Under (Q1)–(Q3), $J = A/\kappa$ is an orthogonal complex structure. The pair (g, J) defines a Hermitian form*

$$h(u, v) = g(u, v) - ig(Ju, v), \tag{20}$$

and therefore a finite complex Hilbert carrier after completion and quotient by any common null directions.

Proof. Because ω is antisymmetric, $g(Au, v) = -g(u, Av)$. Thus A is g-skew-adjoint. The lock gives $J^2 = -I$. Moreover

$$g(Ju, Jv) = -g(u, J^2 v) = g(u, v), \tag{21}$$

so J is g-orthogonal. Equation (20) is conjugate symmetric, complex-linear under multiplication by J, and positive because $h(u, u) = g(u, u) > 0$. Hence it is a Hermitian inner product on the physical quotient. □

Remark 1. *The lock is a physical completion condition, not a theorem of finite loss. If A^2 has unequal negative eigenvalues, one obtains mode-dependent complex scales; if A is degenerate, part of the carrier is phase-blind. Both are testable alternative branches and neither may be silently normalized away.*

9 Unitary transport and the terminal clock

Paper V supplies an observer-relative calibrated duration t_O on a regular ordered leaf. It does not by itself supply quantum dynamics. Under (Q4) and (Q5), define

$$X = JK/U_A. \tag{22}$$

Since K is g-self-adjoint and $[K, J] = 0$, X is g-skew-adjoint and complex-linear. Therefore

$$U(t_O) = e^{t_O X} \tag{23}$$

preserves h. In complex notation this is

$$iU_A \frac{d}{dt_O}|\psi\rangle = H|\psi\rangle, \qquad H = H^{\dagger}. \tag{24}$$

The identification $U_A = \hbar$ is fixed by the standard quantum limit. Paper VI explains the role of the constant: it is the common action conversion between ordered phase transport and calibrated terminal time. It does not derive the numerical SI value of $\hbar$ from the minimal Tau parent.

10 Born and interference terminal

Theorem 6 (Born and interference terminal)**.** *Let* $\rho \geq 0$, $\operatorname{Tr}\rho = 1$ *be an occupied carrier state and let* $\{E_a\}$ *be positive effects with* $\sum_a E_a = I$*. If the source-owned record functional is the normalized positive trace pairing, then*

$$p(a|\rho) = \operatorname{Tr}(\rho E_a) \tag{25}$$

is a normalized probability law. For a rank-one record and coherent alternatives $\psi = \psi_1 + e^{i\phi}\psi_2$*, it contains the cross term*

$$\|\psi\|^2 = \|\psi_1\|^2 + \|\psi_2\|^2 + 2\operatorname{Re}\Big(e^{i\phi}\langle\psi_1, \psi_2\rangle\Big). \tag{26}$$

Proof. Positivity gives $p(a|\rho) \geq 0$, while normalization gives $\sum_a p(a|\rho) = \operatorname{Tr}\rho = 1$. The second statement follows by expanding the Hermitian norm. The phase-sensitive cross term is absent from a classical mixture, which instead has $\rho = \rho_1 + \rho_2$ without off-diagonal coherence. □

In dimensions at least three, noncontextual frame measures are constrained by Gleason-type representation results [5]. We do not use this as a derivation of (Q6): the physical realization of noncontextual positive effects and the source trace remains part of the joint packet. This avoids claiming the Born rule from Hilbert-space notation alone.

11 Record transport and recoverability

The finite Uhlmann geometry developed in Paper III enters here only as a transport condition. A noisy record edge is a CPTP map $\mathcal{C}$. On a declared occupied operator system $\mathcal{S}_{\text{occ}}$, exact recovery means there exists CPTP $\mathcal{R}$ such that

$$(\mathcal{R} \circ \mathcal{C})(X) = X, \qquad X \in \mathcal{S}_{\text{occ}}. \tag{27}$$

Then the distinguishability used by the terminal is preserved on that domain; outside it, contractivity is the correct generic statement. Uhlmann amplitudes provide the purification geometry [4], while Stinespring dilation and Kraus representations supply the standard channel framework [1, 2].

Exact and approximate recoverability, information–disturbance and operator-system correction are standard quantum-information structures; they are not claimed as Tau innovations. Their role here is diagnostic: they state what the observer descent must preserve if the same body-derived packet is to support a complete logical quantum terminal. The specifically Tau-dependent content is the proposed common source of that recoverable operator system and the body-action/Uhlmann identification, neither of which follows from channel theory alone.

This paper distinguishes four domains that are often conflated: a state family, a linear operator system, a code subspace and a correctable algebra. Exact recovery of one does not automatically imply recovery of the full matrix algebra. Approximate recovery requires an explicit error ledger, for example $\|\mathcal{RC} - \mathrm{id}\|_\diamond \leq \eta$; no exact finite-action identity is claimed from this inequality alone.

12 Subsystem-factorization boundary

The occupied P3 block is a full matrix algebra $\mathcal{A}_Q \simeq M_8(\mathbb{C})$, hence it admits abstract realizations as $M_2(\mathbb{C})^{\otimes 3}$. This dimension statement does not identify which observables belong to which physical subsystem. The full S4 effect interval and S5 spectral instruments are likewise invariant under inner automorphisms and therefore do not by themselves break this ambiguity. Accordingly, the tensor-product algebra below is a finite operator-algebraic selector relative to declared source subalgebras, not a derivation of spatial subsystems from matrix dimension alone.

Proposition 2 (One-record factorization no-go)**.** *Fixing one pointer algebra, or even one factor $\mathcal{A}_1 \simeq M_2(\mathbb{C})$ inside $M_8(\mathbb{C})$, does not select a three-factor tensor structure. The commutant of $\mathcal{A}_1$ is $M_4(\mathbb{C})$ and retains multiple candidate decompositions into two $M_2(\mathbb{C})$ factors relative to the frozen source descriptor.*

Proof. In the standard representation let $\mathcal{A}_1 = M_2 \otimes I_2 \otimes I_2$. Its commutant is $I_2 \otimes M_4$. Conjugation inside this commutant moves candidate second and third factors while fixing $\mathcal{A}_1$. More strongly, the diagonal entangling family $U_\theta = \exp(i\theta Z \otimes Z \otimes I)$ fixes every computational-basis pointer projector pointwise while conjugating the first two local Pauli algebras out of their original spans for generic θ. It preserves the normalized trace, the full effect interval and joint generation of M_8. Thus neither one record atlas nor one selected factor removes the ambiguity. □

Theorem 7 (Two-factor selector)**.** *Suppose the source owns two commuting unital subalgebras of $M_8(\mathbb{C})$, denoted by $\mathcal{A}_1$ and $\mathcal{A}_2$. Each is isomorphic to $M_2(\mathbb{C})$, and their multiplication representation is faithful:*

$$C^*(\mathcal{A}_1, \mathcal{A}_2) \simeq M_4(\mathbb{C}). \tag{28}$$

Then

$$\mathcal{A}_3 = C^*(\mathcal{A}_1, \mathcal{A}_2)' \simeq M_2(\mathbb{C}) \tag{29}$$

and the three factors jointly generate $M_8(\mathbb{C})$. The resulting factorization is unique relative to the selected pair, up to local inner automorphisms and permutation of equally typed factors.

Proof. Finite-dimensional representation theory puts the faithful commuting pair in the form $M_2 \otimes M_2 \otimes I_2$. The double-commutant theorem gives Eq. (29); multiplying by its commutant produces $M_2 \otimes M_2 \otimes M_2 = M_8$. Skolem–Noether leaves only inner changes inside the selected simple factors, together with permutations allowed by their source types. □

The finite audit gives factor dimensions $(4, 4, 4)$, joint complex operator rank 64, and commutant complex dimensions $(16, 4, 1)$ after selecting one, two and three factors. The present ROOT edge can anchor a record relation, but the current source ledger does not yet provide the required commuting M_2 pair. Consequently this theorem localizes rather than closes the physical composition problem. The P3 roles, ROOT edge and complete S4/S5 atlas construct a single-system quantum packet but do not yet derive subsystem locality.

There is a more source-proximal form of the selector. Let an *oriented P3 Clifford flag* be an orthogonal decomposition of the seven-dimensional P3 role space into three oriented two-planes and one oriented line. For an adapted Clifford frame $\gamma_1, \ldots, \gamma_7$, put

$$z_j = -i\gamma_{2j-1}\gamma_{2j}, \qquad x_j = \left(\prod_{k<j} z_k\right)\gamma_{2j-1}, \qquad y_j = \left(\prod_{k<j} z_k\right)\gamma_{2j}. \tag{30}$$

The Clifford relations imply that (x_j, y_j, z_j) generates one M_2 factor, different triples commute, and their product generates M_8. Therefore a source-owned Clifford flag is sufficient to realize the two-factor selector and its third commutant. The present isotropic P3 metric, typed role inventory and one occupied ROOT edge do not select such a flag. Before orientation and factor ordering there are already $7\,(6-1)!! = 105$ choices of a residual line and a pairing of the other six axes. A complete ROOT-edge matching or another independently required source polarization could select one; mere role naming cannot.

The tetrahedral seed gives a tempting but incomplete realization. Its six edge labels possess the canonical partition into three opposite-edge pairs, so they reproduce the combinatorial shape of the required flag. However, the existing equivariant CHDF pair map from four seed roles into those six labels has rank four. Its two-dimensional [22] complement is precisely the sector annihilated by the current scalar-plus-quadrupole curvature map, and the root-stabilizer supplies no nonzero direct load into it. Therefore the present tetrahedral body cannot provide six independent metric directions and cannot realize the Clifford flag. This is a rank obstruction, not a failure of the opposite-edge pairing idea. The flag must instead come from a richer base–seed response that physically activates the missing doublet, or from an independently source-owned post-body P3 polarization.

This also joins the factorization problem to the central-block problem. In an irreducible $\mathrm{Cl}_7(\mathbb{C})$ block the seventh Clifford generator is, up to the chosen chirality convention, the product $z_1 z_2 z_3$. Thus a full six-edge flag together with one source-owned chirality sign selects both an M_8 summand and its three M_2 factors. A minimal positive completion can retain the established rank-four CHDF equilibrium while realizing the missing [22] doublet as a zero-background, positive-Hessian fluctuation sector. It does not require a nonzero symmetry-breaking load. What remains unproved is that the physical base assigns this doublet the same edge action unit and selects one chirality. These are two clauses of one finite packet, not new terminal choices.

Proposition 3 (Tetrahedral-packet independence)**.** *The current base–seed axioms do not entail either positive* [22] *stiffness or a* Cl_7 *central character. Consequently they do not entail the six-edge/chirality tensor packet.*

Proof. On the established rank-four CHDF image, append a [22] coordinate w with quadratic term $\lambda\langle w, w\rangle/2$. Both $\lambda = 0$ and every $\lambda > 0$ preserve the same equilibrium $w = 0$, the same CHDF loading and all current S_4 covariance data, while only the latter supplies a positive two-dimensional fluctuation metric. Hence current data do not force $\lambda > 0$. Independently, the two irreducible M_8 representations differ by the sign of the central Clifford volume element. The occupied seed sign $\nu = +1$ exists before this odd Clifford element is introduced and is not a character of its center; both central signs are compatible with the current seed data. Thus neither required clause is entailed. □

Conversely, a common-unit term $\mathcal{A}_*\langle w, w\rangle/2$ together with a source character $\chi(\Omega) = \epsilon \in \{\pm 1\}$ is sufficient. The first completes the six-edge metric, the opposite-edge matching supplies the three planes, and the second chooses the simple block and fixes $\gamma_7 = \epsilon z_1 z_2 z_3$. Equation (30) then constructs the tensor factors. This is a finite sufficient common-source completion relative to the frozen tetrahedral matching. A positive [22] metric and a central character are necessary data types for this route, but factorization alone would not fix the metric coefficient to $\mathcal{A}_*$. Physical occupation of both clauses remains open.

This enrichment is deliberately narrow. The tetrahedral edge representation decomposes as $[4] \oplus [31] \oplus [22]$, while the established CHDF image is exactly $[4] \oplus [31]$. Hence [22] is the unique missing source type and its dimension is exactly the rank deficit. Moreover, Schur's lemma makes the space of S_4-invariant symmetric Hessians on this irreducible doublet one-dimensional:

$$K_{W_2} = \lambda_{W_2} I_{W_2}. \tag{31}$$

The audit verifies this invariant-Hessian dimension directly over all 24 tetrahedral permutations. Thus the proposed body enrichment introduces no arbitrary tensor or response function. It remains inadmissible as physical input until the parent fixes $\lambda_{W_2} > 0$ and its action unit without terminal fitting. The $\lambda_{W_2} = 0$ model is the preregistered rejection control.

There is a source-derived route that avoids adjoining an independent doublet. The tetrahedral decoder identifies the six-edge carrier with one scalar plus the five-dimensional STF representation. The compact seed supplies a rank-three STF orbit, while source-frozen base occurrence transports generate

$$V_{\mathrm{gen}} = \mathrm{span}_p\, \rho_2(P_p) W_{\mathrm{seed}}. \tag{32}$$

The existing covariant body solve is positive on this generated carrier, and full edge completion is equivalent to $\dim V_{\mathrm{gen}} = 5$. An explicit two-frame witness reaches rank five, so the missing doublet can arise from the same seed realized in distinct base-owned relational frames, without new seed payload or a quantum-terminal fit. Physical closure still requires the actual base holonomy to have rank five and the inherited edge Gram to make the three opposite-edge planes mutually orthogonal. Thus occurrence transport derives a conditional W_2 source, not yet the complete Clifford flag.

The remaining Gram test is negative for the naive opposite-edge flag. Under the natural Frobenius metric, opposite edges have zero inner product within a pair, but every entry between two distinct pair planes is $1/4$. The edge-Gram spectrum is $(1/2, 1/2, 1, 1, 1, 2)$, and pair-common/difference coordinates give the tetrahedral decomposition $([4] \oplus [22]) \oplus [31]$, not three orthogonal two-planes. Thus occurrence transport can source the full carrier and its positive Hessian without selecting quantum subsystems. A further source-owned polarization must pair the three relative directions with an orthonormal frame of the scalar-plus-doublet sector; it cannot be chosen by terminal scoring.

A physical root makes this final polarization type-correct. Restriction from S_4 to the root stabilizer S_3 sends both three-dimensional sides to $\mathbf{1} \oplus W_2$. Schur's lemma therefore gives every equivariant mixed Gram in the finite form

$$C_{\mathrm{root}} = c_1 P_{\mathbf{1}} + c_2 P_{W_2}. \tag{33}$$

If $c_1 c_2 \neq 0$, its polar part is the unique metric-compatible rooted isometry and selects the three two-planes without separate gains. Root covariance alone permits either coefficient to vanish and permits relative sign controls, so it does not prove the selector. The remaining source audit is exactly whether the frozen body Hessian owns the nondegenerate mixed block in Eq. (33).

Proposition 4 (Current-body rooted-polarization no-go)**.** *The frozen rank-four CHDF body Hessian cannot own a nondegenerate mixed Gram of the form in Eq.* (33)*. More precisely, its represented pair image contains the complete pair-difference module* [31] *but only the scalar line of the pair-common module* $[4] \oplus [22]$*. Consequently its rooted common/difference mixed block has rank at most one and necessarily* $c_2 = 0$*.*

Proof. The exact pair decomposition is $\mathbb{R}^6_{\mathrm{edge}} = [4] \oplus [31] \oplus [22]$, whereas the current CHDF image is $\operatorname{Im} \Phi_{\mathrm{sum}} = [4] \oplus [31]$. Pair differences are precisely the three-dimensional [31] summand and pair commons are $[4] \oplus [22]$. Hence the intersections of the current image with the difference and common subspaces have dimensions three and one, respectively; the common [22] doublet is absent. After restriction to the rooted S_3, the difference side is $\mathbf{1} \oplus W_2$, but the represented common side contains only $\mathbf{1}$. Schur equivariance therefore leaves at most the scalar-to-scalar coefficient c_1 and forces $c_2 = 0$. The numerical audit checks the intersection dimensions directly. The known nonzero CHDF common/root-standard entry $-\sqrt{5}/8$ lies in this retained scalar block and does not restore the missing W_2 leg. □

Occurrence transport can make the missing [22] carrier available, but rank five alone does not create its mixed Hessian with the rooted W_2 inside [31]. Moreover, an unrooted S_4-equivariant Hessian cannot mix the inequivalent irreducibles [31] and [22]. The route can therefore reopen only through a source-owned rooted mixed incidence whose two W_2 legs, common action unit, adjoint and nondegeneracy descend from the same body functional. This is a localized source obligation, not permission to append another terminal-selected body module.

The hub already contains a conditional realization of precisely this obligation. A source-owned tetrahedral root orders every opposite pair as root-incident/opposite. On the six-edge carrier define the root involution

$$R_* = \operatorname{diag}(1, -1, 1, -1, 1, -1). \tag{34}$$

It exchanges the orthonormal pair-difference and pair-common frames and commutes with the root stabilizer S_3. Hence its restriction $J_{\mathrm{root}} : V_{\mathrm{diff}} \to V_{\mathrm{common}}$ is a canonical rooted isometry, not an independently chosen connector.

Proposition 5 (Conditional common-potential closure)**.** *Assume that occurrence completion supplies the full six-edge carrier, the source owns the root in Eq.* (34)*, and the adopted TC-CVP1* *common strictly convex parent potential restricts to the typed common/difference pair. Then its normalized quadratic germ is*

$$\mathcal{A}_{\mathrm{root}} = \frac{\mu_0}{2}\|c - J_{\mathrm{root}} d\|^2, \qquad \mu_0 > 0, \tag{35}$$

and the mixed Hessian is $H_{cd} = -\mu_0 J_{\mathrm{root}}$*. Consequently* $c_1 = c_2 = -\mu_0 \neq 0$*, so its polar part selects the required rooted tensor polarization without a block-dependent gain.*

Proof. In every opposite pair, multiplication by $(1,-1)$ sends the normalized difference vector $(1,-1)/\sqrt{2}$ to the normalized common vector $(1,1)/\sqrt{2}$. Permutations fixing the root permute the three pairs and therefore commute with R_*. Thus J_{root} is orthogonal and S_3-equivariant on the complete three-dimensional modules. Differentiating Eq. (35) gives unit pure blocks and the stated mixed block. Its restriction to both irreducible rooted summands is the same nonzero scalar multiple of the identity. The mixed block therefore has rank three and its polar factor is $-J_{\mathrm{root}}$. □

This result reuses the existing rooted-occurrence and common-convex-potential branches; it does not prove their physical joint occupation. The current rank-four CHDF Hessian still obeys Proposition 4. The occurrence results sharpen this boundary: MBC-P1/ACT-M1 already supplies the universal four-certificate support and its source root in the adopted working completion. If those certificates are physical body certificates, TC-CVP1 uniquely supplies the typed joint quadratic germ. Thus root ownership and the mixed block are not two further independent selectors.

Proposition 6 (Common-source reduction and rank-five independence)**.** *Within the adopted MBC-P1/ACT-M1 plus TC-CVP1 completion, tensor-polarization closure is equivalent, apart from central chirality, to*

$$\dim \operatorname{span}_p \rho_2(P_p) W_{\mathrm{seed}} = 5, \tag{36}$$

where the P_p *are source-frozen occurrence-frame transports. MBC root ownership and the typed common-potential law do not by themselves imply Eq. (36).*

Proof. MBC-P1/ACT-M1 identifies every occurrence with a source-basic embedding of the same four-certificate cell and selects its pointed root. The physical certificate semantics of TC-CVP1 then gives $H_{cd} = -\mu_0 J_{\mathrm{root}}$ whenever the complete pair carrier is present, so no additional connector or gain remains. The compact tetrahedral seed orbit spans a rank-three subspace $W_{\mathrm{seed}} \subset \mathrm{STF}_5$. If all P_p preserve this subspace, including the aligned choice $P_p = I$, the generated span still has dimension three while the same root, support, unit and TC-CVP1 law remain available. A generic second source-frozen frame can instead enlarge the span to dimension five. These two realizations are an exact equal-root/equal-law counterpair, proving independence. Once the rank is five, Proposition 5 makes the mixed block nondegenerate, proving the stated equivalence inside the adopted completion. □

Unrestricted physical closure therefore requires one remaining common-occupation fact: the physical base–seed occurrence connection must satisfy Eq. (36). The independently required central chirality is still not selected. The split-potential control has the same pure blocks and zero mixed Hessian, so physical realization remains falsifiable rather than definitional.

The minimum physical completion can nevertheless be closed without another free coefficient. Adopt the following recovery requirement:

> **OHC-P1 (irreducible occurrence-holonomy completion).** On the occupied anisotropic carrier, the source-frozen occurrence holonomy has no nonzero proper invariant subspace of STF_5.

This is the weakest invariant-subspace condition that excludes a permanently hidden tensor sector. It is admitted as a 4D-constrained working-completion axiom because the target quantum theory requires the full three-factor carrier; it is not presented as a theorem of the minimal Tau kernel or as an empirical fact.

Theorem 8 (Terminal rank-five closure in the physical MVP). *Let $W_{\text{seed}} \subset \mathrm{STF}_5$ be the nonzero tetrahedral rank-three seed image. Under OHC-P1,*

$$\operatorname{span}_p \rho_2(P_p) W_{\text{seed}} = \mathrm{STF}_5\,. \tag{37}$$

Consequently MBC-P1/ACT-M1, OHC-P1 and the typed TC-CVP1 restriction select the nondegenerate rooted tensor polarization of Proposition 5. No further continuous tensor selector remains in the physical MVP.

Proof. The orbit span is nonzero and invariant under the occurrence-holonomy group: applying another holonomy element only permutes the subspaces included in its definition. OHC-P1 admits no nonzero proper invariant subspace, so the orbit span must be all of STF_5. The rank is therefore five. Proposition 5 then gives a rank-three mixed Hessian with $c_1 = c_2 = -\mu_0 \neq 0$, which completes the rooted polarization. □

The aligned rank-three countermodel remains scientifically useful: it proves that OHC-P1 adds physical content and gives the exact rejection control. The remaining physical question is now whether observations support the OHC-P1 working completion, not whether another formal prerequisite can be introduced.

13 Tensor composition and no-signalling

Theorem 9 (Local CPTP no-signalling). *Assume (Q7) and a bipartite state ρ_{AB} on $\mathcal{H}_A \otimes \mathcal{H}_B$. For every CPTP map $\mathcal{E}_B$,*

$$\mathrm{Tr}_B[(I_A \otimes \mathcal{E}_B)(\rho_{AB})] = \mathrm{Tr}_B\, \rho_{AB}. \tag{38}$$

Thus deterministic local operations on B cannot change the statistics of all effects measured on A.

Proof. Let $\mathcal{E}_B(X) = \sum_k M_k X M_k^\dagger$ with $\sum_k M_k^\dagger M_k = I_B$. For every effect E_A,

$$\mathrm{Tr}[(E_A \otimes I_B)(I_A \otimes \mathcal{E}_B)(\rho_{AB})] = \sum_k \mathrm{Tr}[(E_A \otimes M_k^\dagger M_k)\rho_{AB}] \tag{39}$$

$$= \mathrm{Tr}[(E_A \otimes I_B)\rho_{AB}]. \tag{40}$$

Equality for all E_A proves Eq. (38). □

A selective instrument outcome is trace-non-increasing. Its normalized conditional remote state may change, but the outcome probability must be communicated classically before it can be used. Summing over outcomes restores the CPTP map and Eq. (38). Figure 3 separates these two statements.

Entanglement is therefore available once the tensor carrier is selected: it is nonseparability of ρ_{AB}, not hidden classical correlation. What remains open is why the physical parent selects this tensor composition and its occupied entangled states. The no-signalling theorem is conditional on that composition and cannot be used to derive it retrospectively.

14 Measurement, decoherence and the classical limit

14.1 Quantum instrument

A measurement terminal is represented by an instrument $\{\mathcal{I}_a\}_a$, where each $\mathcal{I}_a$ is completely positive and trace-non-increasing, and $\sum_a \mathcal{I}_a$ is CPTP. Then

$$p_a = \mathrm{Tr}\,\mathcal{I}_a(\rho), \qquad \rho_a = \mathcal{I}_a(\rho)/p_a. \tag{41}$$

This is a consistent terminal record law. It is not yet a proof of a physical parent collapse. The unconditioned state follows the deterministic channel; the conditioned state records the selected outcome.

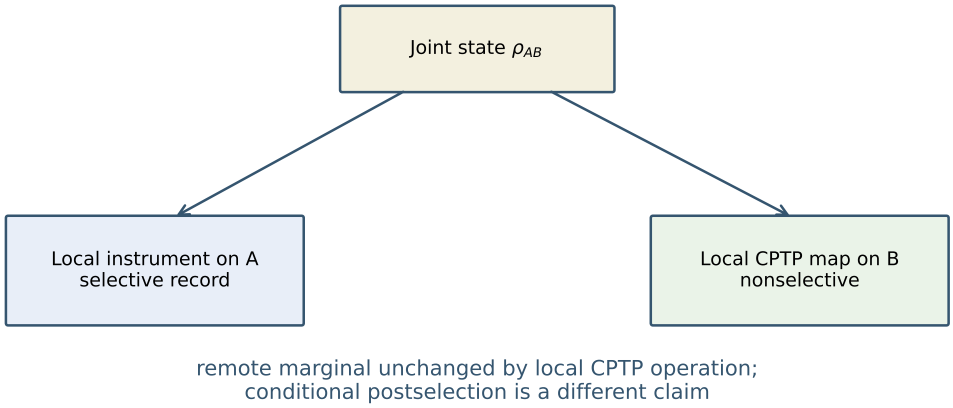

Figure 3: Local deterministic operations preserve the remote marginal. Selective outcomes update conditional states, but the nonselective sum and classical outcome ledger prevent superluminal signalling.

14.2 Decoherence is not outcome selection

For a preferred pointer algebra, a dephasing map removes off-diagonal phase witnesses while preserving populations. Markovian semigroups may be written in Lindblad form [3]. Decoherence explains why interference becomes inaccessible to a coarse observer and why stable records can behave classically. It does not select one realized outcome from the ensemble.

14.3 Classical regime

The classical limit used here requires all of the following:

(C1) suppression of phase-sensitive effects on the occupied record algebra;

(C2) redundant compatible records across accessible observer fragments;

(C3) a stable approximately commuting pointer algebra;

(C4) terminal resolutions coarse compared with the action-phase scale;

(C5) recovery errors below the declared record tolerance.

The limit is therefore observer-relative but not arbitrary: it is constrained by the occupied body relations and by independently testable record stability.

15 Three distinct entropies

Information loss in the descent defines a coarse-graining entropy, for example the log measure of a stable fibre or the Shannon entropy of unresolved classes. This quantity is not automatically the von Neumann entropy

$$S_{\mathrm{vN}}(\rho) = -\operatorname{Tr}(\rho \log \rho), \tag{42}$$

and neither is automatically thermodynamic entropy. The identifications require additional claims:

- coarse-graining entropy equals S_{vN} only when the quantum state and its spectral probabilities are the sufficient observer record;

- thermodynamic entropy requires a macroscopic state space, conserved loads and an equilibrium or typicality argument;
- entropy production requires irreversible effective dynamics or loss to an inaccessible sector, not merely a many-to-one notation.

Recovery theory makes this distinction operational. Exact sufficient recovery can preserve relative entropy [7]; correctable codes can preserve the logical record even through a noisy physical carrier [6]. Hence information loss in one representation need not be physical destruction of the occupied quantum information.

16 Planck action scale

Equation (24) shows what the Planck constant does in this completion. The parent packet supplies one action unit U_A shared by phase and clock descent. Matching the terminal law to quantum mechanics identifies

$$U_A = \hbar. \tag{43}$$

This gives a structural interpretation: $\hbar$ is the action resolution that converts source-owned phase transport into the observer's terminal clock and energy units. It is not proved to be a universal smallest parent action, and the numerical SI value is not predicted. A first-principles result would require a source normalization tied independently to the base–seed action and to operational SI standards.

16.1 Downstream particle and RG handoff

Technical Paper VII-C owns the scalar-15/type-II source preimage, its class-relative mass/Planck ratio and the exact functional-RG evaluation law. Paper VI does not rederive those results. It uses them only to delimit the field-theory handoff: a selected particle branch may share the source action unit while its occupation, component beta functions and SI calibration remain separate physical obligations. The RG regulator is an analysis scheme, not the observer-source quantizer Q_{OS} and not a quantum-readout mechanism.

17 Quantum field theory recovery and frontier

The finite carrier does not by itself yield QFT, but the theory hub already contains a stronger conditional field packet than this paper previously recorded. Assume that the body descent supplies a globally hyperbolic metric and a Green-hyperbolic free operator in the standard locally covariant framework [10]. Its retarded-minus-advanced propagator has causal support, so spacelike-supported observables commute. On a stationary globally hyperbolic background, assume a positive self-adjoint time-translation generator with no obstructing zero modes and the regularity conditions required by the ground-state construction. The resulting compatible pure quasifree covariance is then Hadamard [11]. Positivity of a generic time-dependent occupied operator alone would not establish this conclusion. The existing graph-normal field/occupation block can then add a smooth coherent excitation without changing the Hadamard singularity class.

For local interaction functionals, renormalized time-ordered products obeying causal factorization generate relative S-matrices and spacelike-commuting interacting local algebras order by order [12, 13]. Thus the physical MVP conditionally admits the standard architecture of perturbative QFT on curved spacetime, including microcausality and the time-slice interpretation. These QFT results are imported conditional limits, not new Tau derivations. The Tau-specific question is whether one body-derived source packet supplies their metric, state and local-interaction inputs without independent terminal choices. The paired bilocal control shows why locality must remain an explicit source property: the same free reduct admits nonlocal interactions.

This is not a nonperturbative construction and does not determine the interaction density, couplings, renormalization conditions, spin-statistics, gauge/BRST sector or Standard Model spectrum. A generic time-dependent body also need not select a unique vacuum. Quantum stress and its stochastic second cumulant can source a classical body-derived metric in the joint functional, but a path integral over metrics, its gauge complex and a physical inner product are not derived. The current minimum branch is therefore quantum matter/readout on a classical body-derived geometry, with stochastic backreaction allowed; metric quantization remains undecided.

18 Standard quantum recovery and Tau-specific discriminator

The recovery result and the novelty claim must be kept separate. Under the frozen physical MVP, the following standard structures are recovered:

(R1) positive normalized density operators and POVMs;

(R2) the trace Born rule and coherent interference;

(R3) self-adjoint unitary transport with action unit $U_A = \hbar$;

(R4) tensor composition, singlet correlations and the Tsirelson value $|S_{\mathrm{CHSH}}| = 2\sqrt{2}$ on the selected two-qubit packet [14];

(R5) invariance of a remote marginal under local nonselective CPTP maps;

(R6) free algebraic QFT and the perturbative causal construction under the field assumptions above.

These are necessary compatibility results, not Tau-specific deviations.

The proposed Tau discriminator concerns two quantities that would have to be reconstructed by independent routes. It is not yet an executable prediction, because no audited carrier supplies the required independent action-side observable. For one predeclared occupied record pair define

$$x_A := \frac{\mathcal{A}_X^{PC}}{\mathcal{A}_*}, \qquad x_Q := d^2_{U,\mathrm{chord}}(\rho_0, \rho_1) = 2\left(1 - \sqrt{F(\rho_0, \rho_1)}\right). \tag{44}$$

The common-Gram completion predicts

$$\boxed{x_A = x_Q.} \tag{45}$$

Standard quantum mechanics determines x_Q from state tomography but has no independent morphological action variable x_A and therefore does not imply Eq. (45). This cross-domain identity, rather than a modification of the Born rule, is the primary conditional Tau discriminator.

The prediction remains quantitative under certified approximate recovery. If a record map $\mathcal{C}$ has a recovery $\mathcal{R}$ satisfying $\sup_\rho \|\mathcal{R}\mathcal{C}(\rho) - \rho\|_1 \le \eta$ on the frozen state family, contractivity and the Bures triangle inequality give

$$0 \le x_A - x_Q^{\mathrm{out}} \le x_A - \left[\max\{0, \sqrt{x_A} - 2\sqrt{\eta}\}\right]^2. \tag{46}$$

Thus recovery uncertainty broadens the equality into a precomputable interval; it is not an adjustable fit.

Writing $x_A = \epsilon_O^2$, the common-Gram identity gives the Bures angle $D_B = 2\arcsin(\epsilon_O/2)$. For a constant energy spread under a declared self-adjoint unitary protocol, the standard quantum speed limit then yields

$$\delta t_{\min,O} \ge \frac{2\hbar}{\Delta H} \arcsin\frac{\epsilon_O}{2} \ge \frac{\hbar\epsilon_O}{\Delta H}. \tag{47}$$

The inverse-energy law and Bures speed limit are standard [9, 15]. Tau Core predicts that their distance input is fixed by an independently measured morphological action separation. A timing violation can reject the occupied completion only after Eq. (45), record hold, recovery quality and ΔH have been certified independently.

Define the preregistered residual

$$\Delta_{AQ} := x_Q^{\text{out}} - x_A. \tag{48}$$

Exact recovery predicts $\Delta_{AQ} = 0$; approximate recovery predicts the one-sided interval in Eq. (46). A statistically significant value outside that interval falsifies the common-Gram physical MVP for the tested carrier. Passing it supports the shared-source identity but does not uniquely establish Tau Core, because another theory could impose the same relation. The model currently predicts no universal numerical ϵ_O or laboratory effect size; those require identification of a physical carrier and independent action reconstruction.

19 Public-data audit

We next ask how far Eq. (45) can be tested without a new laboratory protocol. Two public same-carrier datasets already provide nontrivial controls. Tan et al. measured the quantum metric of one superconducting qubit by both sudden quench and weak periodic drive [16]. Extraction from the public vector figures gives 17 matched $g_{\phi\phi}$ points with Pearson correlation 0.9892, MAE 0.01952 and RMSE 0.02222; nine nearly constant $g_{\theta\theta}$ points give MAE 0.00959. The observed cross-protocol offsets are comparable to published decoherence, finite-time and control scales, so they cannot be identified as a Tau gain.

Yu et al. measured metric components and Berry curvature on an NV-centre carrier using distinct parametric modulations [17]. On the seven non-boundary matched coordinates, the pure-qubit identity

$$|F_{\theta\phi}| = 2\sqrt{g_{\theta\theta}g_{\phi\phi} - g_{\theta\phi}^2} \tag{49}$$

has Pearson correlation 0.9933, MAE 0.01545 and RMSE 0.01577 in the public vector-figure extraction. This independently replicates a cross-readout local quantum geometry on a second platform.

These positive results support the standard quantum-geometric recovery and a local same-carrier connector proxy. They do *not* test the finite Tau identity. Every reconstructed axis is a standard quantum-geometric terminal; none is an independently measured once-counted morphological action $\mathcal{A}_X^{PC}$. The public figures also do not provide a certified finite Uhlmann endpoint pair with the required recovery ledger. Therefore current public data determine neither x_A nor Δ_{AQ}. This is a finite identifiability no-go, rather than a null Tau result: the datasets validate part of the standard-recovery side but cannot confirm or falsify Eq. (45). The frozen points, provenance manifest and recalculation script are included in the reproducibility package.

19.1 Compact audit verdict

A frozen public-carrier census applied five hard gates: same carrier, an x_Q-capable endpoint distance, an action-like observable, an independently once-counted x_A, and an uncertainty/recovery ledger. None of the audited candidates passes the independent-x_A gate. The strongest standard controls confirm ordinary quantum geometry and speed-limit behavior, but they do not supply the cross-domain observable required by Eq. (45). The result is therefore an identifiability no-go, not a negative Tau detection.

The strongest QOR control, the trapped-atom data of Ness et al. [18], is consistent with the standard speed limit: no one-sided violation among 53 public points exceeds 1.96σ (maximum 1.4465σ). The Wen optical reconstruction reproduces the phase control and reveals a standard kernel-morphology amplitude dependence, but the source-frozen Tau curvature candidate is null and the optical action is not an independent Tau body action. These results validate parts of the standard recovery pipeline without evaluating $x_A = x_Q$.

Table 2: Compact public-carrier verdict. "Partial" denotes a model-dependent or same-pipeline proxy. The full candidate-by-candidate ledger is preserved in the technical supplement.

Carrier class	same carrier	x_Q	independent x_A	recovery/errors
QGT and tomography platforms	yes	yes	no	partial
Quantum-speed-limit platforms	yes	yes	no	yes/partial
Work, calorimetry and source-port platforms	yes	partial	no	partial
Optical path-integral data	yes	partial	no	partial

There is also a structural no-go. Tomography, QGT and fidelity are circular as x_A estimators; band curvature and state-space QGT are independent in a finite two-band countermodel; energy and fidelity susceptibilities are different spectral moments; and equal local stiffness does not determine a finite Uhlmann chord or loop holonomy. Consequently the minimum admissible reopening packet is

$$\boxed{\{K_A,\ C_q,\ \nabla^C,\ A_*,\ \text{withheld endpoints, loops and recovery ledger}\}}. \tag{50}$$

The body-side stiffness and its action unit must be frozen before the quantum endpoints are examined.

The internal finite audits sharpen but do not remove this boundary. The inherited CHDF quotient supplies a positive body-side action candidate, and exact recovery selects the identity member inside a bounded depolarizing connector family. Generic CPTP structure does not select that member, and ordinary thermodynamic work is not universally proportional to the Bures chord. Thus the remaining physical question is occupation of one common source packet, not another fit of terminal data.

The complete candidate census, all finite countermodels, the alternative source-port survey, Wen path reconstruction, readiness scores and detailed claim boundaries are retained in the separately compiled *Public-Data and Source-Port Audit for the Tau Core Quantum Terminal*. The scripts and frozen data used by both documents remain in this repository.

20 Finite decision protocol

The theory does not need unlimited data collection to test this packet. It needs one source-frozen carrier family on which the following finite gates are independently reconstructed:

(F1) a stable observer partition and ordered record transitions;

(F2) a positive response metric g and phase curvature ω;

(F3) the lock residual $r_J = \|(g^{-1}\omega)^2 + \kappa^2 I\|$;

(F4) a phase-sensitive interference witness excluded by the classical same-quotient control;

(F5) a normalized positive effect ledger testing Eq. (25);

(F6) a compositional no-signalling test with selective and nonselective controls;

(F7) a recovery/dephasing control separating transport loss from outcome statistics.

(F8) independent same-carrier reconstructions of x_A and x_Q, with the recovery tolerance η frozen before evaluating Eq. (45);

(F9) only after that identity passes, an independent ΔH and timing test of Eq. (47).

The branch is weakened if r_J remains nonzero beyond uncertainty, if the phase witness vanishes while the proposed quantum packet is occupied, if probabilities require context-dependent gains, or if a local deterministic operation changes a remote marginal. Passing these tests establishes the finite terminal packet, not its unique parent origin.

21 Parent-realization theorem and exact boundary

Let $\mathfrak{P}_Q$ denote the complete packet consisting of the positive metric, nondegenerate phase form, compatible complex structure, common action unit, separating positive effects, tensor composition and exact recovery on the occupied logical algebra. The present base–seed reduct does not entail $\mathfrak{P}_Q$. Keep the stabilized body, occupied P3 algebra, normalized positive states and upstream source data fixed, and consider

$$\mathcal{C}_\gamma(\rho) = \gamma\rho + (1-\gamma)I/2, \qquad 0 \le \gamma \le 1. \tag{51}$$

Both $\gamma = 1$ and $\gamma = 1/2$ are CPTP descents of the same upstream packet. Orthogonal-state trace distance becomes γ. If a CPTP left inverse existed on the full logical algebra, contractivity of that inverse would require $1 \le \gamma$. Hence $\gamma = 1$ realizes exact recovery and the bounded A8b connector, whereas $\gamma = 1/2$ does not. The aligned and irreducible occurrence transports provide an independent equal-reduct pair with tensor-orbit ranks three and five. Therefore

$$B_\tau^{\mathrm{phys}} + s_U \not\Longrightarrow \mathfrak{P}_Q \tag{52}$$

under the current axioms.

There is nevertheless an exact sufficient source criterion. Call one occupied once-counted parent representation source-faithful and protected when it is simultaneously: (i) action-isometric with the common Hilbert–Schmidt/Uhlmann unit; (ii) metric–symplectic; (iii) separating on the occupied P3 operator system; (iv) monoidal through two faithful commuting $M_2(\mathbb{C})$ factors; and (v) exactly recoverable on the designated logical algebra, with loss confined to its sourced realization complement. For such an SFPR_Q, the preceding theorems give respectively the complex structure, GNS/Born pairing, complete effect atlas, tensor locality, exact A8b member and inherited action unit. Thus

$$\mathrm{SFPR}_Q \Longrightarrow \mathfrak{P}_Q. \tag{53}$$

This is a representation theorem, not an existence theorem. The remaining remaining unrestricted input is physical base–seed occupation of one SFPR_Q; declaring it would close the MVP conditionally, while deriving it requires a stronger parent action or independent physical evidence.

The recovery component has a sharper physical selector. Let $\mathcal{C}_O^c$ be a complementary channel of observer descent. If, on an isolated occupied logical algebra,

$$\mathcal{C}_O^c(\rho_L) = \tau_G \quad \text{for every } \rho_L, \tag{54}$$

then the inaccessible realization complement contains no logical-state information. The finite-dimensional information–disturbance/correctability criterion [6] implies a CPTP left inverse of $\mathcal{C}_O$ on the complete logical algebra. At the same minimal input and output dimension the channel is unitary conjugation. This condition is stronger than stable classical recording: complete dephasing preserves the trace distance of $|0\rangle, |1\rangle$ but maps the orthogonal phase pair $|+\rangle, |-\rangle$ to the same state.

A parent-side sufficient realization of Eq. (54) is a source-frozen isolated split $TM_{\mathrm{occ}} = L \oplus G$ satisfying

$$P_L H_{\mathrm{parent}} P_G = 0, \qquad \ell_{\mathrm{parent}} = \ell_L \oplus \ell_G, qquad U_{\mathrm{parent}} = U_L \otimes U_G, qquad \rho_{LG} = \rho_L \otimes \tau_G. \tag{55}$$

Thus morphological isolation means absence of a mixed source incidence able to export logical distinguishability into G, not absence of the surrounding body. The observed unitary limit of isolated quantum systems requires this law operationally. It is therefore a 4D-constrained isolated-scope selector, not yet a first-principles consequence of the current base–seed action. Turning on the mixed block gives the approximate-recovery regime rather than a contradiction.

21.1 Disjoint-source valuation and the isolation theorem

This source-locality proposal should not be confused with standard cluster decomposition or with algebraic-QFT locality. Those are terminal statements about correlations or commuting observable algebras. The valuation below is a stronger parent-side composition axiom for disconnected occupied source incidences; the paper tests its consequences without claiming that standard locality selects it.

The block-zero condition in Eq. (55) follows from one finite parent composition law. Let the occupied source hypergraph split into disconnected components and require the once-counted parent action to be a normalized disjoint-union valuation,

$$\mathcal{A}(\varnothing) = 0, \qquad \mathcal{A}(X \sqcup Y) = \mathcal{A}(X) + \mathcal{A}(Y) \tag{56}$$

whenever no occupied primitive source incidence meets both X and Y. Induction on the finite connected-component decomposition gives

$$\mathcal{A}(H_{\text{occ}}) = \sum_{c} \mathcal{A}(H_c). \tag{57}$$

Twice differentiating Eq. (57) yields

$$P_c H_{\text{parent}} P_d = 0 \qquad (c \neq d). \tag{58}$$

Equations (56)–(58) therefore prove only the action and Hessian parts of morphological isolation. Exact logical recovery additionally requires the source load, generator and occupied state to factor as stated explicitly in Eq. (55), or an independent proof of the logical non-leakage condition (54). A vanishing mixed Hessian alone does not exclude higher-order, global-constraint or initial-correlation leakage.

The valuation is not implied by once-counting, positivity, source freeze and internal component symmetry alone. A positive parent Hessian may contain

$$\epsilon \, \langle z_L, \mathbf{1} \rangle \langle z_G, \mathbf{1} \rangle, \qquad \epsilon \neq 0, \tag{59}$$

while preserving the two internal permutation symmetries. Equation (59) gives $P_L H_{\text{parent}} P_G \neq 0$ and violates Eq. (56). The valuation is therefore a real composition law, not a restatement of the weaker source packet.

If a physical hyperedge crosses the $L|G$ partition, its second variation may generate precisely such a mixed block. Logical information can then enter G, allowing entanglement, local decoherence and nonexact recovery while the joint parent state remains represented. Exact protection is consequently an isolated-scope theorem, not a universal ban on interaction.

21.2 Operational-to-parent two-jet lifting theorem and no-go

The finite valuation defect $\mathcal{A}(X \sqcup Y) - \mathcal{A}(X) - \mathcal{A}(Y)$ is a scalar and cannot be acted on by a terminal Jacobian. The operationally accessible local object is instead its mixed second variation. For source variations $u \in E_X$ and $v \in E_Y$, define

$$N_{\mathcal{A}}^{X|Y}(u, v) := \frac{\partial^2}{\partial s \, \partial t} \mathcal{A}\big(X(su) \sqcup Y(tv)\big)\bigg|_{s=t=0} \in E_{\text{cross}} := \mathrm{Bil}(E_X, E_Y; \mathbb{R}). \tag{60}$$

Let $\mathfrak{D}^{(2)}_{\text{prim}} : E_{\text{cross}} \to E^{(2)}_{\text{term}}$ be the stacked, source-frozen second-jet descent of all declared primitive physical readouts. This typed map sends a parent mixed bilinear response to the corresponding family of terminal mixed responses.

Proposition 7 (Joint two-jet faithfulness)**.** *Suppose operationally isolated preparations have vanishing mixed terminal responses,*

$$\mathfrak{D}^{(2)}_{\mathrm{prim}} N^{X|Y}_{\mathcal{A}} = 0, \tag{61}$$

for every admissible pair, and suppose

$$\ker \mathfrak{D}^{(2)}_{\mathrm{prim}} \cap E_{\mathrm{cross}} = \{0\}. \tag{62}$$

Then $N^{X|Y}_{\mathcal{A}} = 0$, equivalently $P_X H_{\mathrm{parent}} P_Y = 0$, on that occupied scope.

Proof. The operational premise places $N^{X|Y}_{\mathcal{A}}$ in $\ker \mathfrak{D}^{(2)}_{\mathrm{prim}}$, while Eq. (60) places it in E_{cross}. Equation (62) makes this intersection trivial. □

This proposition lifts only the local mixed two-jet. It does not recover the finite valuation law (56): that stronger statement requires all mixed finite-response jets to vanish, or an independent global composition axiom. Nor does the proposition alone imply exact recovery, which still requires the remaining factors in Eq. (55).

The faithfulness premise is necessary for *universal identifiability over the admitted cross-defect class.* Add to a positive valuation Hessian a small positive-preserving cross form N such that

$$P_L N P_G \neq 0, \qquad \mathfrak{D}^{(2)}_{\mathrm{prim}} N = 0. \tag{63}$$

The original and modified parents have identical declared primitive two-jet reducts, while only the first has a zero mixed Hessian. Thus no inference valid for the full admitted defect class can establish parent two-jet isolation while a common cross-defect null direction survives. A narrower physical model may exclude the counterform by an additional source law, so faithfulness is not asserted as a logically necessary axiom of every possible completion.

The present series does not yet establish Eq. (62). A global physical $\mathfrak{D}^{(2)}_{\mathrm{prim}}$ cannot yet be formed on one frozen domain because full electric/magnetic/radiative gravity, photon and gauge occupation, non-Abelian/chiral matter, interacting multi-system QFT and their common calibration are not all completed. The local six-dimensional Green packet is complete on its own descriptor, but that local rank statement is not global primitive-readout completeness. Thus the exact status is

$$\begin{gathered} \text{joint two-jet faithfulness} \Longrightarrow \text{zero parent mixed Hessian}, \\ \text{zero parent mixed Hessian} + \text{ global factorization data} \Longrightarrow \text{protection}, \end{gathered} \tag{64}$$

with the first premise physically open and an equal-readout countermodel excluding a stronger universal inference from the current reduct. In particular, the absence of a completed global terminal stack establishes non-demonstrability, not a theorem that the physical kernel is nonzero.

The possibility-space intuition requires one further distinction. Let C^+_x be the forward cone selected by the body order and $K_x > 0$ the local body Hessian. If the local admissibility law is reversal symmetric, then $v \mapsto -v$ is an isometry between the forward and backward quadratic acceptance bodies. Orientation and stiffness alone therefore cannot enlarge the future. Let R_- and R_+ instead denote the source-owned restriction maps imposed on backward and forward compatible extensions. Their fibres are

$$F^-_x = \ker R_-, \qquad F^+_x = \ker R_+. \tag{65}$$

A larger future possibility space means a larger dimension or inherited Hessian volume of F^+_x, as can occur when seed and current-body data fix backward compatibility more strongly than forward completion.

This asymmetry becomes exact when backward extension must also reproduce the observer's already stable records. Let R_{rec} be the corresponding record-agreement map on the current cut and set

$$R_- = (R_+, R_{\mathrm{rec}}). \tag{66}$$

Then

$$F_x^- = kerR_+ \cap \ker R_{\rm rec} \subseteq F_x^+, \qquad \dim F_x^+ - \dim F_x^- = \operatorname{rank}\Big(R_{\rm rec}|_{F_x^+}\Big). \tag{67}$$

The second identity is rank–nullity applied to the restricted record map. Consequently the future fibre is strictly larger exactly when a stable record constraint removes at least one otherwise future-compatible direction. At fixed resolution ϵ, r independent removed directions can increase the operational number of resolved future cells relative to past cells by a factor scaling as ϵ^{-r} on a normalized bounded chart. This cell count, rather than a comparison of volumes in different dimensions, is the precise sense in which the future can be much more "fluid."

No stored parent history is used: $R_{\rm rec}$ is a simultaneous stability condition in the observer's current readout quotient. Nor does the observer access actual future outcomes. Equation (67) establishes excess compatible directions, not their quantum coherence; the phase form and future–ROOT bridge remain necessary for that promotion.

The word "larger" does not mean unbounded. The admissible local future body is

$$\mathcal{E}_x^+(\Lambda) = F_x^+ \cap C_x^+ \cap \mathcal{D}_M \cap \{v : g_+(v,v) \leq \Lambda\}, \tag{68}$$

where $\mathcal{D}_M$ contains the remaining morphology constraints. If F_x^+ is finite dimensional, $g_+ > 0$, $\Lambda < \infty$ and the causal and morphology constraints are closed, Eq. (68) is compact and has finite Hessian volume. A finite observer resolution then produces only finitely many stable readout cells. The future is more fluid only relative to the record-constrained past; the morphological body still sets its directions, stiffness, boundary and accessible resolution.

The available concrete body quotient also gives an exact dimensional boundary. The CHDF leaf Hessian is positive on one radial and two tangent directions, hence has real rank three. If that leaf alone were identified with F_x^+, every linear bridge to the real rank-four ROOT qubit would have rank at most three; an antisymmetric phase form on the leaf has even rank at most two. Thus the CHDF-leaf-only identification cannot realize the full ROOT carrier. This is not a no-go for the enriched body: the P3/ROOT packet already supplies the conditional rank-four target, but no source-frozen body-first matrix R_+ is currently available on that same domain. The numerical rank of the mixed future–ROOT block is consequently undefined, not zero. The next finite construction is an enriched body/P3 restriction retaining a four-real-dimensional non-gauge, metric-visible sector before the ROOT terminal is applied.

The inclusion $i_+ : F_x^+ \hookrightarrow T_x M_\tau$ induces

$$g_+ = i_+^* K_x > 0. \tag{69}$$

This is a classical possibility geometry. A symmetric Hessian cannot itself produce phase. If the same source supplies a quotient-basic curvature ω_+ with

$$(g_+^{-1}\omega_+)^2 = -\kappa^2 I, \tag{70}$$

then $J_+ = g_+^{-1}\omega_+/\kappa$ gives the complex carrier and the existing coherent positive-effect theorems apply. The observer accesses a current descriptor of the compatible extension fibre, not an already realized future outcome. Operational future-blindness and no-signalling are therefore preserved. Physical realization of R_+, its rank advantage and the common source of ω_+ remain open.

The last open phrase can be sharpened by using the variational boundary structure of one parent action rather than postulating an unrelated phase law. Suppose an ordered differentiable parent action has first variation

$$\delta\mathcal{A}_M = \langle E(M), \delta M\rangle + \int_{\Sigma_O} \theta_M(\delta M), \tag{71}$$

where Σ_O is the current observer cut. On compatible linearized future solutions, the bulk second variation induces the Jacobi metric g_+, while the antisymmetrized boundary variation induces

$$\omega_+(\delta_1, \delta_2) = \delta_1\theta_M(\delta_2) - \delta_2\theta_M(\delta_1). \tag{72}$$

Thus metric and phase have one once-counted variational owner. After removing the gauge/null kernel, assume $g_+ > 0$ and nondegenerate ω_+. With $A = g_+^{-1}\omega_+$, polar decomposition gives

$$J_+ = A(-A^2)^{-1/2}, \qquad J_+^2 = -I. \tag{73}$$

This conditionally derives the complex carrier without an independent phase gain. It does not imply the stronger scalar lock $A^2 = -\kappa^2 I$: that requires all polar mode scales to coincide. The finite certificate with $g_+ = \mathrm{diag}(1, 1, 2, 2)$ and two equal canonical phase blocks has a valid polar complex structure but mode scales $(1, 1, 1/2, 1/2)$. It is therefore an explicit counterexample to deriving one universal action unit from common variational ownership alone. Physical occupation, quotient descent and the equal-mode-scale law remain open.

The equal-mode-scale condition itself has a finite representation-theoretic closure. Let a parent symmetry group G act irreducibly on the occupied real future fibre and preserve both g_+ and ω_+. Then $A = g_+^{-1}\omega_+$ commutes with G, and so does the positive self-adjoint polar operator $|A| = (-A^2)^{1/2}$. Every eigenspace of $|A|$ is G-invariant. Irreducibility therefore forces

$$|A| = \kappa I, \qquad A^2 = -\kappa^2 I. \tag{74}$$

This is the required scalar lock, not an additional fitted gain. A real four-dimensional quaternionic certificate has a one-dimensional symmetric commutant and zero lock residual. By contrast, a reducible sum of two phase planes retains scales 1 and 1/2. Thus irreducibility is essential. The current base–seed law has not yet been shown to occupy this irreducible future-fibre orbit, and Eq. (74) fixes only relative mode scales, not the absolute value or SI calibration of κ.

No new irreducible carrier is required for this step. The occupied logical ROOT qubit is a complex two-dimensional, hence real four-dimensional, irreducible $SU(2)$ module. Let

$$C_{FQ} : F_x^+ \longrightarrow \mathcal{H}_{\mathrm{ROOT}}^{\mathbb{R}} \tag{75}$$

be a source-owned bijective intertwiner preserving the metric and phase forms. Pulling the ROOT action back through C_{FQ} makes F_x^+ irreducible, so Eq. (74) follows. The finite realified Pauli certificate has bridge rank four, zero intertwining residual and a one-dimensional symmetric commutant.

Equation (75) is not selected by the current reduct. Keeping the stabilized body and complete internal ROOT packet fixed admits both $C_{FQ} = 0$ and a full-rank equivariant C_{FQ}. Thus the bridge theorem reduces physical occupation to one typed source incidence but does not derive that incidence from the present base–seed law.

The source of C_{FQ} can be localized further. Decompose the boundary second variation of Eq. (71) into future and ROOT blocks,

$$\omega_{\partial M} = \begin{pmatrix} \omega_F & \beta_{FQ} \\ -\beta_{FQ}^T & \omega_Q \end{pmatrix}. \tag{76}$$

If F_x^+ and $\mathcal{H}_{\mathrm{ROOT}}^{\mathbb{R}}$ are equivalent irreducible parent modules and β_{FQ} is a nonzero equivariant intertwiner, then its kernel and image are invariant. Irreducibility forces a zero kernel and full image, so β_{FQ} itself supplies a full-rank C_{FQ}. Separate bijectivity is not an independent assumption.

The diagonal restrictions ω_F and ω_Q do not activate this block. A finite antisymmetric family with the same two diagonal phase forms admits both $\beta_{FQ} = 0$ and $\beta_{FQ} = \lambda I$; for $0 < |\lambda| < 1$ the latter joint form is nondegenerate and its bridge has rank four. Hence common variational ownership alone still does not force the mixed incidence. This block is distinct from the forbidden logical–environment mixing in Eq. (55): β_{FQ} loads the protected logical carrier from the body possibility fibre, while morphological isolation prevents that logical information from leaking into the inaccessible realization complement.

The previously derived BRAC term does not by itself supply this boundary block. Its morphology-dependent potential is zeroth order, so it has a generically nonzero mixed bulk

Hessian on an occupied field but contributes no independent covariant symplectic potential. The required source instead occurs in the covariant kinetic term of the selected joint completion (JCSEL, BRDC and EBRP). On an observer cut its canonical momentum is

$$\pi_\Phi = Z_\Phi \sqrt{h}\, n^\mu \partial_\mu \Phi. \tag{77}$$

For a future body variation δ_F and field variation $\delta\Phi$, the mixed boundary coefficient contains

$$\beta_{FQ}(\delta_F, \delta\Phi) \supset \delta_F\Big(Z_\Phi \sqrt{h}\, n^\mu\Big)\, \partial_\mu \Phi\, \delta\Phi. \tag{78}$$

SBTCL595's nonneutral EBRP branch supplies a nonzero occupied positive-frequency field momentum. Hence every non-gauge future mode that changes the densitized normal kinetic metric makes Eq. (78) nonzero. Covariance makes the block equivariant; on the equivalent irreducible future/ROOT sector the preceding Schur argument promotes it to the full bridge.

This proves activation inside the selected covariant working completion. It does not prove that the physical base–seed parent selects JCSEL–BRDC–EBRP, that the realized body is nonneutral, or that every future-fibre direction is metric-visible. The unrestricted physical statement therefore remains conditional, while the internal completion question is closed.

22 What is proved and what remains open

Table 3: Claim ledger. "Conditional theorem" means mathematically derived from the explicitly frozen packet, not established as an unrestricted parent-law realization with physical occupation.

Claim	Status and condition	Class
Loss alone implies quantum mechanics	False; classical and quantum completions share one finite quotient	No-go theorem
Complex carrier	Follows from positive g, nondegenerate ω and $A^2 = -\kappa^2 I$	Conditional theorem
Unitary transport	Follows from Paper-V time, one action unit and commuting self-adjoint K	Conditional theorem
Born and interference	Follows from positive trace effects and coherent composition	Conditional theorem
No-signalling	Follows for local CPTP maps after tensor composition is fixed	Conditional theorem
Measurement	Quantum instruments are consistent; physical single-outcome selection is not derived	Open physics
Classical limit	Decohered commuting records give a controlled effective limit; not a fundamental collapse	Conditional limit
Planck scale	$U_A = \hbar$ by standard-limit calibration; SI value not derived	Identification
Downstream scalar-15 scale and interacting RG	owned by Technical Paper VII-C; imported only as a field-theory boundary	not part of the quantum terminal derivation
Current reduct selects the packet	False: Eq. (51) gives an equal-upstream-reduct counterfamily	Exact non-entailment theorem
Joint source criterion	One occupied source-faithful protected representation implies the complete packet by Eq. (53); its physical occupation is open	Conditional representation theorem
Isolation selector	Logical non-leakage in Eq. (54) forces exact recovery; the factorized parent law in Eq. (55) is sufficient but not yet derived from the base–seed action	Theorem plus physical-MVP selector

Continued on next page

Claim	Status and condition	Class
Disjoint-source valuation	Exact finite additivity forces a block-diagonal Hessian, but the Hessian statement alone does not force global dynamics, state factorization or recovery	Conditional composition theorem
Operational parent lift	Jointly faithful typed second-jet descent lifts vanishing terminal mixed responses to a zero parent mixed Hessian; it does not derive finite valuation or recovery by itself, and global faithfulness is not yet demonstrable	Conditional local theorem plus identifiability no-go
Future-extension carrier	Boundary-fibre asymmetry can enlarge the forward possibility space and the restricted Hessian supplies its metric; phase lock and physical fibre selection remain conditional	Theorem/no-go plus conditional completion
Record-conditioned fluidity	Stable backward record agreement gives $F_x^- \subseteq F_x^+$ with dimension gap equal to the restricted record rank; coherence of the excess directions is not automatic	Exact rank theorem
Bounded future body	Positive restricted Hessian, finite action cutoff and closed body/causal constraints make the admissible future compact with finite observer-resolved cell count	Conditional compactness theorem
Common-action phase source	Bulk and boundary variations of one parent action conditionally supply g_+ and ω_+; polar decomposition gives J_+, but equal polar scales and one universal action unit do not follow	Conditional theorem plus finite no-go
Irreducible mode-scale lock	A common parent symmetry preserving g_+ and ω_+ forces $\lvert g_+^{-1}\omega_+\rvert = \kappa I$ on an irreducible occupied future fibre; physical orbit occupation and absolute calibration remain open	Conditional Schur-rigidity theorem
Future–ROOT bridge	A bijective source-owned metric–phase intertwiner to the logical ROOT qubit transfers its real irreducibility and closes the relative scale lock; zero/full-rank equal-reduct completions prove current nonselection	Conditional bridge theorem plus exact non-entailment
Mixed boundary source	A nonzero equivariant future–ROOT boundary block is automatically full rank between the equivalent irreducible carriers; identical diagonal packets admit zero/nonzero blocks, so physical activation is open	Conditional Schur theorem plus exact counterfamily
Kinetic activation	BRAC bulk mixing is the wrong type, but the selected JCSEL–BRDC–EBRP kinetic momentum gives $\delta_F \pi_\Phi \neq 0$ for occupied non-null metric-visible future modes; unrestricted completion realization remains open	Conditional physical-completion theorem
Single-system common source	Derived inside the enriched MVP plus adopted P3 handoff; unchanged-parent realization remains open	Conditional source theorem
P3 handoff selector	Separating records with one common scalar quadratic action force the operator-square and graph-normal law; physical occupation remains open	Conditional source theorem
P3 record rank	Primitive and binary families have kernels 56 and 35; grade-≤ 3 closure has rank 64; physical ternary occupation remains open	Exact finite rank theorem
Spectral record atlas	Canonical binary effects of all grade-≤ 3 words are separating; Frobenius pointer copying alone has rank eight; physical spectral closure follows inside the full S4/S5 MVP	Conditional source theorem/no-go
Spectral closure selects P3 backwards	False: the involutions used by the closure already assume P3HAND-D1	Circularity no-go

Continued on next page

Claim ledger (continued)

Claim	Status and condition	Class
Unique tensor factorization	Not selected by the simple matrix algebra and normalized trace alone; closed in the physical MVP by MBC/ACT, OHC-P1 and the typed TC-CVP1 polarization	Conditional completion theorem
Tensor selector	A regular event incidence plus exact minimal-qubit recovery selects one factor axis; canonical expectation fixes its M_4 complement, while recovery error η bounds off-factor weight by η	Conditional theorem
Clifford-flag realization	Three oriented orthogonal P3 two-planes plus one line construct the commuting factors by Eq. (30); current source data do not select the flag	Conditional source theorem
Tetrahedral flag candidate	Opposite edges give the correct three-pair combinatorics, but the current CHDF pair image has rank four and cannot carry the required six independent directions	Exact inherited rank no-go
Six-edge completion	A positive zero-background [22] fluctuation doublet completes the edge carrier and its common M_8 quantum algebra	Conditional selector
Central chirality	Both odd Cl_7 extensions restrict to the same six-edge M_8 terminal; the sign is invisible here and remains open only for a parity-odd/Weyl-sensitive extension	Exact terminal-blindness theorem
Physical realization of the completion	Zero/positive [22] stiffness obeys current source data; unrestricted six-edge occupation is not entailed	Exact independence theorem
Anti-overfitting status	The missing module is uniquely [22] and its invariant Hessian has one scalar coefficient; positivity and the common unit remain source-selection tests	Structural candidate
Occurrence-generated W_2	OHC-P1 forces every nonzero seed orbit to span the full rank-five STF carrier; unrestricted OHC-P1 occupation remains open	Physical-MVP theorem/empirical boundary
Natural tetrahedral Gram	Opposite pairs are internally orthogonal but the three pair planes have nonzero cross blocks; the natural split is $3 + (1 + 2)$, not $2 + 2 + 2$	Exact no-go
Rooted polarization	Both sides restrict to $1 \oplus W_2$; a nondegenerate source mixed Gram selects a unique polar isometry, but root covariance alone does not force it	Conditional selector/no-go
Perturbative interacting QFT	Local causal factorization and Hadamard preparation recover the curved-spacetime architecture conditionally; interaction content and nonperturbative existence remain open	Conditional standard limit
Quantum geometry	Matter/readout is quantum on a classical body-derived metric with possible stochastic backreaction; metric quantization is undecided	Open richer completion
Tau-specific discriminator	Independent same-carrier reconstructions would obey $x_A = x_Q$, broadened by Eq. (46) under certified recovery error, only on the occupied bounded sector $0 \leq x_A \leq 2$; existence of a matching carrier is proved but its physical realization remains open; exact full-code recovery selects the matching depolarizing member and approximate recovery gives a frozen factor-mixing tolerance; physical occupation of the joint packet remains open; no audited public carrier makes the two axes independently observable, and the QOR timing bound is downstream	Proposed conditional discriminator; not yet executable

The former physical open arrow is now resolved negatively for the current reduct:

$$B_\tau^{\mathrm{phys}} + s_U \not\Longrightarrow \mathfrak{P}_Q. \tag{79}$$

The surviving positive question is narrower: does the physical base–seed body occupy the single SFPR_Q representation characterized above? Independent post-hoc choices would reproduce quantum notation but not a unified Tau explanation. Physical occupation, rather than another internal connector calculation, is now the primary blocker.

23 Conclusion

Observer-relative information loss has a precise but limited role. It creates stable operational classes and can make microscopic discreteness coexist with macroscopic continuity. It does not create quantum mechanics on its own. A genuine quantum terminal requires a phase-capable metric–symplectic source packet, positive effects and coherent composition. Once these are frozen, the complex carrier, unitary transport, Born/interference law and local CPTP no-signalling follow with no terminal fit. OHC-P1 closes the continuous tensor-polarization selector inside the physical MVP, and the inherited Hadamard/causal-factorization packet extends the standard recovery to perturbative curved-spacetime QFT under explicit field assumptions. Within the resulting tensor packet, exact minimal-qubit recovery, regular observer-event incidence and canonical conditional expectation jointly fix the contextual logical factor and its realization complement. Approximate recovery supplies a quantitative stability bound. The unselected central Clifford sign does not affect this even quantum terminal.

The paper's distinguishing proposal is not a modified Born rule or an unexplained timing delay. It is the common-Gram identity $x_A = x_Q$, with a recovery-error envelope and a downstream QOR speed limit. The public-data audit shows that this proposal is not yet executable because no carrier makes x_A and x_Q independently observable. Paper VI is therefore a conditional quantum foundation with an operationally specified future falsification target, not a completed quantum theory of the physical universe. Unrestricted physical occupation, carrier identification, effect size, nonperturbative QFT, metric quantization and the Standard Model remain open.

The scalar-15 and RG packet owned by Technical Paper VII-C sharpens the field-theory handoff without changing this verdict. It does not prove that the unrestricted base–seed parent occupies that branch, and it does not turn an RG cutoff into the mechanism of observer-relative quantum readout.

The isolation result is correspondingly conditional but sharper than a bare block-zero assumption. Disjoint-source valuation forces action additivity and a zero mixed Hessian, while a crossing hyperedge gives the local interaction boundary. A typed, jointly faithful second-jet descent can lift vanishing operational mixed responses back only to that parent two-jet statement. Exact recovery still requires factorized loads, generators and occupied states, and finite valuation remains a separate global source law. Since the known terminal families do not yet form one complete physical stacked map, even the local unrestricted parent lift remains unproved rather than silently assumed.

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